

Tafila Technical University
IT and Communication College
Department of Computer Science

Course: AI Programming
Final Project

Instructions

- Students may work individually or in groups of up to **3 students maximum**.
- I expect and require that **all group members participate equally** in the implementation and presentation of the project.
- Each group will have **10 minutes** to present and demonstrate their project.
- All students must fully understand their code and be able to explain any part of the implementation.
- The use of AI tools such as ChatGPT or GitHub Copilot is allowed only for assistance and learning purposes.
- I will check for:
 - AI-generated coding patterns
 - Similarity with online solutions
 - Understanding of the code during the presentation
 - Ability to modify and explain the implementation live
- Any project copied entirely from online sources or generated using AI without understanding may receive a zero.

Required Python Concepts

Each project **must use** the following concepts covered during the semester:

- Data Types
- Selection Statements (**if, elif, else**)
- Loops (**for, while**)
- Functions

- Strings
- Lists
- Sets
- Dictionaries
- Lambda Expressions
- Iterators and Functional Programming Concepts:
 - `map()`
 - `filter()`
 - `zip()`
 - `enumerate()`

Important: Projects that do not clearly demonstrate the use of the required concepts may lose marks.

Project Ideas

1. Smart Student Grade Manager

- Store student grades
- Calculate averages
- Find highest and lowest grades
- Generate statistics

2. Password Strength Analyzer

- Check password complexity
- Detect weak passwords
- Suggest stronger passwords

3. Text Analyzer System

- Count words and characters
- Find repeated words
- Detect longest word

4. Smart Contact Management System

- Add, delete, and update contacts
- Search contacts

5. Mini E-Commerce System

- Product management
- Shopping cart
- Checkout system

6. Student Attendance System

- Record attendance
- Generate attendance reports

7. AI-Based Quiz Game

- Random questions
- Score calculation
- Difficulty levels

8. Word Frequency Analyzer

- Analyze text frequency
- Display most common words

9. Smart File Organizer

- Organize files by type
- Detect duplicate files

10. Python Banking System

- Deposit and withdraw
- Transfer money
- Transaction history

11. Data Cleaning System

- Remove invalid data
- Use `map()`, `filter()`, and `lambda`

12. Student Data Transformer

- Transform names and grades
- Filter failed students
- Use `zip()` and `enumerate()`

13. Functional Calculator

- Build calculator using lambda expressions
- Use dictionaries for operations

14. Smart Inventory Management System

- Product tracking

- Low-stock alerts

15. **Employee Payroll System**

- Salary calculations
- Tax calculations
- Employee filtering

16. **Email Validator**

- Validate email addresses
- Detect invalid patterns

17. **Text Encryption System**

- Encrypt and decrypt messages
- Caesar cipher implementation

18. **Social Media Comment Filter**

- Remove offensive words
- Filter unwanted comments

19. **Hashtag Analyzer**

- Analyze hashtags
- Find trending tags

20. **Username Generator**

- Generate usernames automatically
- Use random values and strings

21. **Sudoku Validator**

- Validate Sudoku board
- Detect invalid rows or columns

22. **AI Tic-Tac-Toe Game**

- Human vs computer gameplay
- Smart move generation

23. **Smart Schedule Planner**

- Create study schedules
- Organize tasks automatically

24. **Expense Tracker**

- Track expenses

- Categorize spending
- Generate reports

25. Library Management System

- Book management
- Borrow and return system
- Search functionality

Suggested Deliverables

Each project should include:

- Python source code
- Documentation/report
- Screenshots
- Presentation slides
- Project demonstration

Evaluation Criteria

Component	Weight
Correctness and Functionality	30%
Use of Python Concepts	25%
Problem Solving and Logic	15%
Code Organization and Readability	10%
Creativity and Additional Features	10%
Presentation and Understanding	10%